CONTRAT API WORKCENTER - COCKPIT OBSIDIA / X108 - V1 COMPLETE

Document 2/4 - Contrat API WorkCenter - Pre-build - Version complete

CONTRAT API WORKCENTER - COCKPIT OBSIDIA / X108 - V1 COMPLETE

Document 2/4 de la suite pre-build. Ce support convertit le cahier des charges et la checklist d implementation en contrat API concret pour developpeur. Il ne lance aucun build. Il ne modifie aucun fichier projet existant. Il definit les routes, les payloads, les reponses, les statuts, les erreurs, les tests et les boundaries que le Cockpit / WorkCenter doit respecter.

Ce contrat est volontairement long et operationnel. Son objectif est d empecher l improvisation lors du developpement. Chaque route est reliee a une version, une page cockpit, un ticket, un mode d action, des tests et des interdits.

1. Position du document dans la suite de supports

Les supports valides ou en cours forment une chaine. Le cahier des charges explique quoi construire. La checklist explique dans quel ordre le construire. La matrice relie les pages aux tickets. Le present contrat API definit precisement les routes qui relient le frontend, le backend et les registries.

- Document 1 - Matrice Cahier des charges -> Checklist -> Tickets DEV : relie les pages aux tickets.
- Document 2 - Contrat API WorkCenter : definit les routes, payloads, reponses, statuts, erreurs et tests.
- Document 3 - Object Model / Data Model Registry : definit les objets manipules par les routes.
- Document 4 - Security & Authority Boundary Matrix : definit les droits et interdits de chaque composant.

2. Doctrine API

Le WorkCenter API n est pas un nouveau cerveau. C est une couche de lecture, de preparation, de validation et d audit. Les routes GET lisent. Les routes POST creent des demandes, des candidates, des propositions ou des validations. Aucune route du WorkCenter ne doit rendre une decision souveraine a la place du Kernel X108.

2.1 Regles absolues

- Le Kernel X108 reste la seule autorite de decision ALLOW / HOLD / BLOCK.
- FastAPI reste BRIDGE_ONLY pour les flux decisionnels.
- Le Cockpit reste READONLY_CONTROL_SURFACE ou VALIDATION_SURFACE selon la page.
- Les routes POST du WorkCenter ne sont pas des actions reelles: elles creent des demandes, candidates, preparations ou validations tracees.
- La mutation directe du Kernel depuis le Cockpit ou l API WorkCenter est interdite.
- L evolution du Kernel passe seulement par Kernel Change Protocol, qui fait l objet d une demande auditee, jamais d une mutation directe.
- Graphiti est considere readonly tant qu un protocole de write explicite n est pas cree et valide.
- La memoire canonique exige validation humaine.
- Les preuves doivent afficher leur vrai statut: test, freeze, code, documented only, present not run, env blocked ou failed.
- Les secrets ne doivent jamais etre renvoyes par une route WorkCenter.

3. Versioning API

Le contrat API est versionne selon la feuille de route cockpit. Les routes peuvent etre ajoutees progressivement, mais elles doivent respecter les schemas et boundaries des maintenant.

- V0 - Socle de registries et routes readonly de base.
- V1 - Cockpit readonly visible: dashboard, authority, servers, domains, routes.
- V2 - Gouvernance memoire, validation humaine, validation process, session ledger.
- V3 - Proof & Replay Lab, Freeze Factory, manifests, compare freezes.
- V4 - Brody Education, Resource Library, Terrain Branching, Agent Center, Knowledge Center.
- V5 - Coding Sandbox, Multimodal Workbench, Governance Policies, Kernel Change Protocol view, Audit Center, Acceptance Matrix.

4. Enveloppe standard de reponse

Toutes les routes WorkCenter doivent pouvoir etre enveloppees dans un format standard. Ce format rend les reponses lisibles par le frontend et auditables. Il doit etre utilise au minimum pour les nouvelles routes V0 a V5.

```
{
  "ok": true,
  "route": "/api/workcenter/example",
  "method": "GET",
  "version": "V1",
  "mode": "READONLY",
  "truth_status": "CONFIRMED_BY_CODE",
  "decision_authority": "KX108_ONLY",
  "cockpit_role": "READONLY_CONTROL_SURFACE",
  "boundaries": {
    "decision_authority": "KX108_ONLY",
    "cockpit_role": "READONLY_CONTROL_SURFACE",
    "allowed_to_decide": false,
    "emits_act": false,
    "emits_verdict": false,
    "memory_write": false,
    "graphiti_write": false,
    "kernel_mutation": false,
    "neo4j_write": false,
    "real_action": false,
    "canonical_write": false
  },
  "data": {},
  "warnings": [],
  "errors": [],
  "trace": {
    "trace_id": "trace_example",
    "timestamp": "2026-06-17T12:00:00Z"
  }
}
```

5. Enums canoniques

5.1 TruthStatus

```
CONFIRMED_BY_RUNTIME
CONFIRMED_BY_TEST
CONFIRMED_BY_FREEZE
CONFIRMED_BY_CODE
PRESENT_NOT_CONNECTED
DOCUMENTED_ONLY
LEGACY
BLOCKED
FAILED
ENV_BLOCKED
UNKNOWN
```

5.2 MemoryStatus

```
RAW_CAPTURE
SESSION_SCRATCH
CANDIDATE
NEEDS_HUMAN_REVIEW
```

```
NEEDS_PROCESS_REVIEW
PROCESS_VALIDATED
HUMAN_VALIDATED
PROMOTED_TO_CANON
PROMOTED_TO_TRAINING
PROMOTED_TO_OPERATIONAL
PROMOTED_TO_DOMAIN
PROMOTED_TO_PROOF
REJECTED
SUPERSEDED
DEPRECATED
FORBIDDEN_SECRET
```

5.3 ValidationStatus

```
PENDING
APPROVED
REJECTED
CHANGES_REQUESTED
EXPIRED
BLOCKED_BY_POLICY
```

5.4 ProofStatus

```
PROVEN_BY_TEST
PROVEN_BY_PRIOR_AUDIT
PROVEN_BY_FREEZE
PRESENT_NOT_RUN
ENV_BLOCKED
FAILED
UNKNOWN
```

5.5 FreezeStatus

```
DRAFT
CANDIDATE
VALIDATION_REQUIRED
READY_FOR_SMOKE
SMOKE_RUNNING
SMOKE_PASS
SMOKE_FAIL
FROZEN
REJECTED
BLOCKED_BY_POLICY
```

5.6 AuthorityRole

```
SOVEREIGN_DECISION_ENGINE
BRIDGE_ONLY
READONLY_EDUCATION
FROZEN_READONLY_MEMORY
ADVISORY_AGGREGATION
READONLY_CONTEXT
CONTROL_SURFACE
HUMAN_VALIDATOR
PROOF_VERIFIER
```

6. Catalogue erreurs standard

400 VALIDATION_ERROR - Payload ou query parameter invalide.

401 AUTH_REQUIRED - Utilisateur non authentifie.

403 FORBIDDEN_BY_BOUNDARY - La route existe mais l action viole une boundary.

403 FORBIDDEN_SECRET_SCOPE - Le fichier, secret ou contenu est interdit.

404 NOT_FOUND - Ressource introuvable.

409 STATUS_CONFLICT - Le statut de la ressource ne permet pas l action demandee.

409 TRUTH_STATUS_DOWNGRADE_BLOCKED - La demande tente de surclasser ou degrader un statut sans preuve.

422 UNSUPPORTED_TARGET - La cible demandee n est pas supportee par la version.

423 VALIDATION_REQUIRED - Une validation humaine ou process est obligatoire.

424 DEPENDENCY_NOT_CONFIRMED - Une dependance n est pas confirmee au niveau requis.

500 INTERNAL_ERROR - Erreur serveur non attendue.

503 SERVICE_UNAVAILABLE - Service indisponible ou source runtime fermee.

7. Index des routes par version

V0

- GET /api/workcenter/status — Foundation / Registries — [V0] [API] [BACKEND] WorkCenter root status
- GET /api/workcenter/metrics — Foundation / Registries — [V0] [API] [BACKEND] Metrics globales
- GET /api/workcenter/registries — Foundation / Registries — [V0] [API] [BACKEND] Index des registries
- GET /api/workcenter/components — Foundation / Registries — [V0] [API] [BACKEND] Composants
- GET /api/workcenter/components/{component_id} — Foundation / Registries — [V0] [API] [BACKEND] Détail composant
- GET /api/workcenter/routes — Foundation / Registries — [V0] [API] [BACKEND] Routes connues
- GET /api/workcenter/routes/detail — Foundation / Registries — [V0] [API] [BACKEND] Détail route
- GET /api/workcenter/boundaries — Foundation / Registries — [V0] [API] [BACKEND] Boundaries globales

V1

- GET /api/workcenter/dashboard — Dashboard global — [V1] [DASHBOARD] [API] Dashboard data
- GET /api/workcenter/decisions/recent — Dashboard global — [V1] [DASHBOARD] [API] Décisions récentes
- GET /api/workcenter/risks/open — Dashboard global — [V1] [DASHBOARD] [API] Risques ouverts
- GET /api/workcenter/authority — Authority Graph — [V1] [AUTHORITY] [API] Graphe autorité
- GET /api/workcenter/authority/forbidden-actions — Authority Graph — [V1] [AUTHORITY] [API] Actions interdites
- GET /api/workcenter/servers — Server Advisory — [V1] [SERVER] [API] Serveurs
- GET /api/workcenter/ports — Server Advisory — [V1] [SERVER] [API] Ports
- GET /api/workcenter/commands/suggested — Server Advisory — [V1] [SERVER] [API] Commandes conseillées
- GET /api/workcenter/domains — Domain Center — [V1] [DOMAIN] [API] Domaines
- GET /api/workcenter/domains/{domain_id} — Domain Center — [V1] [DOMAIN] [API] Détail domaine
- GET /api/workcenter/pages/{page_id}/routes — Routes Registry — [V1] [ROUTES] [API] Routes par page

V2

- GET /api/workcenter/memory/candidates — Memory Governance — [V2] [MEMORY] [API] Candidates mémoire
- GET /api/workcenter/memory/promoted — Memory Governance — [V2] [MEMORY] [API] Memories promues
- GET /api/workcenter/memory/{memory_id} — Memory Governance — [V2] [MEMORY] [API] Détail memory entry
- POST /api/workcenter/memory/candidate — Memory Governance — [V2] [MEMORY] [API] Créer candidate
- POST /api/workcenter/memory/promotion/request — Memory Governance — [V2] [MEMORY] [API] Demande promotion mémoire

- GET /api/workcenter/validation/human — Human Validation — [V2] [VALIDATION] [API] Queue validation humaine
- GET /api/workcenter/validation/human/{validation_id} — Human Validation — [V2] [VALIDATION] [API] Détail validation humaine
- POST /api/workcenter/validation/request — Human Validation — [V2] [VALIDATION] [API] Créer validation request
- POST /api/workcenter/validation/human/{validation_id}/approve — Human Validation — [V2] [VALIDATION] [API] Approve validation
- POST /api/workcenter/validation/human/{validation_id}/reject — Human Validation — [V2] [VALIDATION] [API] Reject validation
- POST /api/workcenter/validation/human/{validation_id}/changes-requested — Human Validation — [V2] [VALIDATION] [API] Changes requested
- GET /api/workcenter/validation/process — Process Validation — [V2] [PROCESS] [API] Queue process validation
- GET /api/workcenter/validation/process/{process_id} — Process Validation — [V2] [PROCESS] [API] Détail process validation
- GET /api/workcenter/session/current — Session Ledger — [V2] [SESSION] [API] Session actuelle
- GET /api/workcenter/sessions — Session Ledger — [V2] [SESSION] [API] Liste sessions
- GET /api/workcenter/sessions/{session_id} — Session Ledger — [V2] [SESSION] [API] Détail session
- POST /api/workcenter/session/close/dry-run — Session Ledger — [V2] [SESSION] [API] Clôture dry-run

V3

- GET /api/workcenter/proofs — Proof & Replay Lab — [V3] [PROOF] [API] Proofs summary
- GET /api/workcenter/proofs/{proof_id} — Proof & Replay Lab — [V3] [PROOF] [API] Détail proof
- POST /api/workcenter/proofs/run/prepare — Proof & Replay Lab — [V3] [PROOF] [API] Préparer proof run
- GET /api/workcenter/replay/decisions — Proof & Replay Lab — [V3] [REPLAY] [API] Replays disponibles
- GET /api/workcenter/replay/decisions/{decision_id} — Proof & Replay Lab — [V3] [REPLAY] [API] Détail replay
- GET /api/workcenter/freezes — Freeze Factory — [V3] [FREEZE] [API] Freezes
- GET /api/workcenter/freezes/{freeze_id} — Freeze Factory — [V3] [FREEZE] [API] Détail freeze
- POST /api/workcenter/freezes/candidate — Freeze Factory — [V3] [FREEZE] [API] Préparer freeze candidate
- GET /api/workcenter/freezes/{freeze_id}/manifest — Freeze Factory — [V3] [FREEZE] [API] Lire manifest
- GET /api/workcenter/freezes/compare — Freeze Factory — [V3] [FREEZE] [API] Comparer freezes

V4

- GET /api/workcenter/brody/training/modules — Brody Education — [V4] [BRODY] [API] Modules training
- GET /api/workcenter/brody/training/modules/{module_id} — Brody Education — [V4] [BRODY] [API] Détail module
- POST /api/workcenter/brody/training/quiz/dry-run — Brody Education — [V4] [BRODY] [API] Quiz dry-run
- GET /api/workcenter/resources — Resource Library — [V4] [RESOURCE] [API] Ressources
- GET /api/workcenter/resources/{resource_id} — Resource Library — [V4] [RESOURCE] [API] Détail ressource
- GET /api/workcenter/terrain/domains — Terrain Branching — [V4] [TERRAIN] [API] Domaines terrain
- GET /api/workcenter/terrain/branches — Terrain Branching — [V4] [TERRAIN] [API] Branches
- POST /api/workcenter/terrain/branches/candidate — Terrain Branching — [V4] [TERRAIN] [API] Créer branche candidate
- GET /api/workcenter/agents — Agent Center — [V4] [AGENT] [API] Agents
- GET /api/workcenter/knowledge/candidates — Knowledge Center — [V4] [KNOWLEDGE] [API] Knowledge candidates

V5

- GET /api/workcenter/sandbox/policy — Coding Sandbox — [V5] [SANDBOX] [API] Policy sandbox
- GET /api/workcenter/sandbox/files/tree — Coding Sandbox — [V5] [SANDBOX] [API] Arborescence readonly
- GET /api/workcenter/sandbox/files/read — Coding Sandbox — [V5] [SANDBOX] [API] Lire fichier readonly
- POST /api/workcenter/sandbox/patch-proposals — Coding Sandbox — [V5] [SANDBOX] [API] Créer patch proposal
- POST /api/workcenter/sandbox/test-proposals — Coding Sandbox — [V5] [SANDBOX] [API] Test proposal

- GET /api/workcenter/multimodal/assets — Multimodal Workbench — [V5] [MULTIMODAL] [API] Assets multimodaux
- POST /api/workcenter/multimodal/extractions/candidate — Multimodal Workbench — [V5] [MULTIMODAL] [API] Créer extraction candidate
- GET /api/workcenter/governance/policies — Governance Policies — [V5] [POLICY] [API] Policies
- GET /api/workcenter/governance/policies/{policy_id} — Governance Policies — [V5] [POLICY] [API] Détail policy
- POST /api/workcenter/kernel/change-requests — Kernel Change Protocol — [V5] [KERNEL_CHANGE] [API] Créer request changement Kernel
- GET /api/workcenter/audit/events — Audit Center — [V5] [AUDIT] [API] Audit events
- GET /api/workcenter/acceptance/criteria — Acceptance Matrix — [V5] [ACCEPTANCE] [API] Acceptance criteria

8. Contrats detailles par route

8.1 GET /api/workcenter/status

Version : V0

Page cockpit : Foundation / Registries

Ticket DEV : [V0] [API] [BACKEND] WorkCenter root status

Mode : READONLY

Objectif

Retourner le statut global du WorkCenter, le role du cockpit, l autorite decisionnelle, les statuts des services et les flags de boundary. Cette route est la premiere route de confiance du cockpit.

Request

```
No request body.
```

Response minimale attendue

```
{
  "cockpit_role": "READONLY_CONTROL_SURFACE",
  "decision_authority": "KX108_ONLY",
  "api": {
    "port": 8000,
    "status": "CONFIRMED_BY_RUNTIME",
    "role": "BRIDGE_ONLY"
  },
  "kernel": {
    "port": 3001,
    "status": "CONFIRMED_BY_FREEZE",
    "role": "SOVEREIGN_DECISION_ENGINE"
  },
  "boundaries": {
    "decision_authority": "KX108_ONLY",
    "cockpit_role": "READONLY_CONTROL_SURFACE",
    "allowed_to_decide": false,
    "emits_act": false,
    "emits_verdict": false,
    "memory_write": false,
    "graphiti_write": false,
    "kernel_mutation": false,
    "neo4j_write": false,
    "real_action": false,
    "canonical_write": false
  }
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_status.py
- tests/test_workcenter_no_decision_flags.py

Boundaries et interdits

- Retourner cockpit_decides=true
- Retourner fallback_decision=true
- Masquer un Kernel indisponible
- Transformer CONFIRMED_BY_FREEZE en CONFIRMED_BY_RUNTIME

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.2 GET /api/workcenter/metrics

Version : V0

Page cockpit : Foundation / Registries

Ticket DEV : [V0] [API] [BACKEND] Metrics globales

Mode : READONLY

Objectif

Retourner les compteurs globaux nécessaires au dashboard: routes, composants, domaines, preuves, freezes, mémoires candidates, validations, warnings et éléments stale.

Request

```
No request body.
```

Response minimale attendue

```
{
  "route_count": 164,
  "component_count": 0,
  "domain_count": 0,
  "proof_count": 57,
  "memory_candidate_count": 0,
  "validation_pending_count": 0,
  "warning_count": 0,
  "stale_count": 0,
  "blocked_count": 0
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_metrics.py

Boundaries et interdits

- Inventer des compteurs si la source manque
- Additionner des sources legacy avec des sources actives sans statut

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.3 GET /api/workcenter/registries

Version : V0

Page cockpit : Foundation / Registries

Ticket DEV : [V0] [API] [BACKEND] Index des registries

Mode : READONLY

Objectif

Lister les registries disponibles, leurs chemins, leurs statuts et leurs dates de vérification. Cette route sert à prouver que l UI lit une source structurée et ne hardcode pas les informations.

Request

```
No request body.
```

Response minimale attendue

```
{
  "items": [
    {
      "id": "components_registry",
      "path": "workcenter/registries/components_registry.json",
      "status": "CONFIRMED_BY_CODE",
      "last_verified": null
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_registries_index.py

Boundaries et interdits

- Retourner une registry inexistante comme active
- Cacher une registry manquante

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.4 GET /api/workcenter/components

Version : V0

Page cockpit : Foundation / Registries

Ticket DEV : [V0] [API] [BACKEND] Composants

Mode : READONLY

Objectif

Retourner tous les composants du système: Kernel, API, Sigma, Brody, Graphiti, CIC, Proof stack, UI, connecteurs, registries et modules de gouvernance.

Request

```
No request body.
```

Response minimale attendue

```
{
  "items": [
    {
      "id": "kernel_x108",
      "label": "Kernel X108",
      "role": "SOVEREIGN_DECISION_ENGINE",
      "truth_status": "CONFIRMED_BY_FREEZE",
      "can_decide": true,
      "can_mutate_from_cockpit": false
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_components.py

Boundaries et interdits

- Déclarer Brody souverain
- Déclarer Graphiti décisionnel
- Déclarer le cockpit souverain

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.5 GET /api/workcenter/components/{component_id}

Version : V0

Page cockpit : Foundation / Registries

Ticket DEV : [V0] [API] [BACKEND] Détail composant

Mode : READONLY

Objectif

Retourner la fiche complète d un composant avec rôle, statut, frontières d autorité, routes liées, tests liés, freezes liés, risques et actions interdites.

Request

```
No request body.
```

Response minimale attendue

```
{
  "id": "kernel_x108",
  "label": "Kernel X108",
  "description": "Moteur souverain ALLOW/HOLD/BLOCK",
  "linked_routes": [
    "POST /kernel/ragnarok"
  ],
  "forbidden_from_cockpit": [
    "direct_mutation",
    "force_allow",
    "fallback_decision"
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_component_detail.py

Boundaries et interdits

- Créer une fiche sans boundary
- Masquer les risques du composant

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.6 GET /api/workcenter/routes

Version : V0

Page cockpit : Foundation / Registries

Ticket DEV : [V0] [API] [BACKEND] Routes connues

Mode : READONLY

Objectif

Retourner le catalogue des routes connues, leur source, leur méthode, leur rôle, leur statut et leur page cockpit associée.

Request

```
No request body.
```

Response minimale attendue

```
{
  "items": [
    {
      "method": "GET",
      "path": "/api/status",
      "role": "STATUS_READ",
      "truth_status": "CONFIRMED_BY_RUNTIME",
      "mutating": false
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_routes_catalog.py

Boundaries et interdits

- Classer une route POST sensible comme readonly sans justification
- Masquer les routes legacy

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.7 GET /api/workcenter/routes/detail

Version : V0

Page cockpit : Foundation / Registries

Ticket DEV : [V0] [API] [BACKEND] Détail route

Mode : READONLY

Objectif

Retourner le détail d une route par méthode et path, en conservant les boundaries, les exemples JSON, les tests et le niveau de vérité.

Request

```
Query parameters: method, path.
```

Response minimale attendue

```
{
  "method": "GET",
  "path": "/api/workcenter/status",
  "truth_status": "CONFIRMED_BY_CODE",
  "request_schema": null,
  "response_schema": "WorkcenterStatus"
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_route_detail.py

Boundaries et interdits

- Retourner un détail sans statut de vérité
- Inventer un schéma non implémenté

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.8 GET /api/workcenter/boundaries

Version : V0

Page cockpit : Foundation / Registries

Ticket DEV : [V0] [API] [BACKEND] Boundaries globales

Mode : READONLY

Objectif

Retourner les flags de boundary globaux qui doivent rester visibles dans toutes les pages: allowed_to_decide, emits_act, emits_verdict, memory_write, graphiti_write, kernel_mutation, neo4j_write, real_action, canonical_write.

Request

```
No request body.
```

Response minimale attendue

```
{
  "decision_authority": "KX108_ONLY",
  "cockpit_role": "READONLY_CONTROL_SURFACE",
  "allowed_to_decide": false,
  "emits_act": false,
  "emits_verdict": false,
  "memory_write": false,
  "graphiti_write": false,
  "kernel_mutation": false,
  "neo4j_write": false,
  "real_action": false,
  "canonical_write": false
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_boundaries.py

Boundaries et interdits

- Retourner true pour une capacité non validée
- Faire varier la boundary par page sans policy

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.9 GET /api/workcenter/dashboard

Version : V1

Page cockpit : Dashboard global

Ticket DEV : [V1] [DASHBOARD] [API] Dashboard data

Mode : READONLY

Objectif

Assembler les données du dashboard sans modifier les sources: services, domaines, proofs, freezes, mémoires, validations, risques et dernières décisions.

Request

```
No request body.
```

Response minimale attendue

```
{
  "summary": {
    "services": {},
    "domains": {},
    "proofs": {},
    "memory": {},
    "validation": {},
    "risks": {}
  },
  "boundaries": {
    "decision_authority": "KX108_ONLY",
    "cockpit_role": "READONLY_CONTROL_SURFACE",
    "allowed_to_decide": false,
    "emits_act": false,
    "emits_verdict": false,
    "memory_write": false,
    "graphiti_write": false,
    "kernel_mutation": false,
    "neo4j_write": false,
    "real_action": false,
    "canonical_write": false
  }
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_dashboard.py

Boundaries et interdits

- Créer une décision
- Lancer un healthcheck destructif
- Masquer un warning

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.10 GET /api/workcenter/decisions/recent

Version : V1

Page cockpit : Dashboard global

Ticket DEV : [V1] [DASHBOARD] [API] Décisions récentes

Mode : READONLY

Objectif

Lister les dernières décisions connues par domaine avec leur source de vérité, sans rejouer ni recalculer la décision.

Request

```
Query parameters: limit, domain_id optional, gate optional.
```

Response minimale attendue

```
{
  "items": [
    {
      "decision_id": "decision_example",
      "domain": "trading",
      "gate": "BLOCK",
      "severity": "S4",
      "reason_code": "CONTRADICTION_THRESHOLD_REACHED",
      "truth_status": "CONFIRMED_BY_FREEZE"
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_recent_decisions.py

Boundaries et interdits

- Recalculer une décision dans le cockpit
- Remplacer une décision manquante par une estimation

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.11 GET /api/workcenter/risks/open

Version : V1

Page cockpit : Dashboard global

Ticket DEV : [V1] [DASHBOARD] [API] Risques ouverts

Mode : READONLY

Objectif

Retourner les risques ouverts issus des audits, pages et registries: preuves non relancées, Neo4j dégradé, services fermés, docs legacy, routes non classées.

Request

```
No request body.
```

Response minimale attendue

```
{
  "items": [
    {
      "id": "risk_tla_not_rerun",
      "severity": "MEDIUM",
      "status": "OPEN",
      "source": "proofs",
      "message": "TLC not rerun in current audit"
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_open_risks.py

Boundaries et interdits

- Masquer un risque connu
- Classer un risque comme fermé sans preuve

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.12 GET /api/workcenter/authority

Version : V1

Page cockpit : Authority Graph

Ticket DEV : [V1] [AUTHORITY] [API] Graphe autorité

Mode : READONLY

Objectif

Retourner les nœuds et arêtes du graphe d autorité: Kernel, FastAPI, Sigma, Brody, Graphiti, CIC, Proof stack, Cockpit, Humain.

Request

```
No request body.
```

Response minimale attendue

```
{
  "nodes": [
    {
      "id": "kernel_x108",
      "label": "Kernel X108",
      "authority_role": "SOVEREIGN_DECISION_ENGINE",
      "can_decide": true
    },
    {
      "id": "cockpit",
      "label": "Cockpit",
      "authority_role": "CONTROL_SURFACE",
      "can_decide": false
    }
  ],
  "edges": [
    {
      "from": "cockpit",
      "to": "fastapi_bridge",
      "kind": "READS"
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_authority.py

Boundaries et interdits

- Créer une arête Cockpit -> Kernel MUTATE
- Créer une arête Brody -> Kernel DECIDE

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.13 GET /api/workcenter/authority/forbidden-actions

Version : V1

Page cockpit : Authority Graph

Ticket DEV : [V1] [AUTHORITY] [API] Actions interdites

Mode : READONLY

Objectif

Lister les actions interdites par composant et par version du cockpit.

Request

```
No request body.
```

Response minimale attendue

```
{
  "items": [
    {
      "component": "cockpit",
      "forbidden_action": "DIRECT_KERNEL_MUTATION",
      "reason": "Requires Kernel Change Protocol"
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_forbidden_actions.py

Boundaries et interdits

- Omettre DIRECT_KERNEL_MUTATION
- Omettre GRAPHITI_WRITE

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.14 GET /api/workcenter/servers

Version : V1

Page cockpit : Server Advisory

Ticket DEV : [V1] [SERVER] [API] Serveurs

Mode : READONLY

Objectif

Afficher les services attendus, ports, statuts runtime ou freeze, warnings et commandes suggérées en copie seule.

Request

```
No request body.
```

Response minimale attendue

```
{
  "items": [
    {
      "service": "api_fastapi",
      "port": 8000,
      "runtime_status": "CONFIRMED_BY_RUNTIME",
      "expected_role": "BRIDGE_ONLY"
    },
    {
      "service": "kernel_x108",
      "port": 3001,
      "runtime_status": "CLOSED",
      "freeze_status": "CONFIRMED_BY_FREEZE"
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_servers.py

Boundaries et interdits

- Démarrer un serveur
- Arrêter un serveur
- Tuer un process

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.15 GET /api/workcenter/ports

Version : V1

Page cockpit : Server Advisory

Ticket DEV : [V1] [SERVER] [API] Ports

Mode : READONLY

Objectif

Retourner les ports attendus, leur état et leur signification dans Obsidia.

Request

```
No request body.
```

Response minimale attendue

```
{
  "items": [
    {
      "port": 8000,
      "service": "FastAPI",
      "status": "LISTENING",
      "truth_status": "CONFIRMED_BY_RUNTIME"
    },
    {
      "port": 3001,
      "service": "Kernel X108",
      "status": "CLOSED",
      "truth_status": "CONFIRMED_BY_FREEZE"
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_ports.py

Boundaries et interdits

- Masquer un port fermé
- Interpréter un port fermé comme erreur si seulement freeze attendu

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.16 GET /api/workcenter/commands/suggested

Version : V1

Page cockpit : Server Advisory

Ticket DEV : [V1] [SERVER] [API] Commandes conseillées

Mode : READONLY

Objectif

Retourner des commandes PowerShell ou bash à copier, sans exécution automatique.

Request

```
No request body.
```

Response minimale attendue

```
{
  "items": [
    {
      "id": "check_api_status",
      "shell": "powershell",
      "command": "Invoke-RestMethod http://127.0.0.1:8000/api/status",
      "mode": "COPY_ONLY"
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_suggested_commands.py

Boundaries et interdits

- Exécuter la commande côté serveur
- Inclure des secrets
- Inclure une commande destructive

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.17 GET /api/workcenter/domains

Version : V1

Page cockpit : Domain Center

Ticket DEV : [V1] [DOMAIN] [API] Domaines

Mode : READONLY

Objectif

Retourner les domaines connus et leurs statuts: Bank, Trading, GPS/Aviation/Defense, Ecom, CIC, NCP, Scraping readonly, futurs domaines.

Request

```
No request body.
```

Response minimale attendue

```
{
  "items": [
    {
      "domain_id": "bank",
      "label": "Bank",
      "status": "CONFIRMED_BY_FREEZE",
      "api_route": "/api/live/kernel/adapters/bank"
    },
    {
      "domain_id": "ecom",
      "label": "Ecom",
      "status": "PRESENT_NOT_CONNECTED"
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_domains.py

Boundaries et interdits

- Marquer Ecom runtime si seulement présent code
- Permettre action réelle depuis domaine

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.18 GET /api/workcenter/domains/{domain_id}

Version : V1

Page cockpit : Domain Center

Ticket DEV : [V1] [DOMAIN] [API] Détail domaine

Mode : READONLY

Objectif

Retourner la fiche d un domaine avec routes, connecteurs, payloads, métriques risque, derniers replays et interdits.

Request

```
No request body.
```

Response minimale attendue

```
{
  "domain_id": "trading",
  "status": "CONFIRMED_BY_FREEZE",
  "last_decision": {
    "gate": "BLOCK",
    "severity": "S4"
  },
  "forbidden_modes": [
    "DIRECT_ACT",
    "DOMAIN_DECIDES_ALONE"
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_domain_detail.py

Boundaries et interdits

- Omettre forbidden_modes
- Créer un mode REAL_ACTION_WITHOUT_X108

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.19 GET /api/workcenter/pages/{page_id}/routes

Version : V1

Page cockpit : Routes Registry

Ticket DEV : [V1] [ROUTES] [API] Routes par page

Mode : READONLY

Objectif

Retourner toutes les routes liées à une page cockpit afin que le dev sache quoi implémenter et tester.

Request

```
No request body.
```

Response minimale attendue

```
{
  "page_id": "dashboard_global",
  "routes": [
    "GET /api/workcenter/status",
    "GET /api/workcenter/dashboard"
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_page_routes.py

Boundaries et interdits

- Laisser une page sans route source
- Hardcoder des données frontend

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.20 GET /api/workcenter/memory/candidates

Version : V2

Page cockpit : Memory Governance

Ticket DEV : [V2] [MEMORY] [API] Candidates mémoire

Mode : READONLY

Objectif

Lister les memory candidates avec source, statut, cible, validation requise, raisons et possibilités d action.

Request

```
No request body.
```

Response minimale attendue

```
{
  "items": [
    {
      "id": "mem_candidate_001",
      "source": "session",
      "status": "NEEDS_HUMAN_REVIEW",
      "target": "CANONICAL_MEMORY",
      "validation_required": "HUMAN",
      "can_promote": false,
      "reason": "Canon promotion requires human validation"
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_memory_candidates.py

Boundaries et interdits

- Promotion automatique
- Écriture Graphiti
- Écriture durable silencieuse

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.21 GET /api/workcenter/memory/promoted

Version : V2

Page cockpit : Memory Governance

Ticket DEV : [V2] [MEMORY] [API] Memories promues

Mode : READONLY

Objectif

Lister les mémoires déjà promues par catégorie: training, operational, domain, proof, canon.

Request

```
No request body.
```

Response minimale attendue

```
{
  "items": [
    {
      "id": "mem_001",
      "type": "TRAINING_MEMORY",
      "status": "PROMOTED_TO_TRAINING",
      "validated_by": "human_or_process",
      "source_refs": []
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_promoted_memory.py

Boundaries et interdits

- Retourner une mémoire sans validation source
- Confondre training et canon

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.22 GET /api/workcenter/memory/{memory_id}

Version : V2

Page cockpit : Memory Governance

Ticket DEV : [V2] [MEMORY] [API] Détail memory entry

Mode : READONLY

Objectif

Afficher une mémoire, ses sources, validations, usages, risques, historique et statut de vérité.

Request

```
No request body.
```

Response minimale attendue

```
{
  "id": "mem_candidate_001",
  "status": "NEEDS_HUMAN_REVIEW",
  "source_refs": [],
  "promotion_history": [],
  "risk": "CANON_PROMOTION_REQUIRES_HUMAN"
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_memory_detail.py

Boundaries et interdits

- Afficher un secret interdit en clair
- Masquer la source

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.23 POST /api/workcenter/memory/candidate

Version : V2

Page cockpit : Memory Governance

Ticket DEV : [V2] [MEMORY] [API] Créer candidate

Mode : CONTROLLED_CREATE_REQUEST

Objectif

Créer une memory candidate en mode demande, sans promotion durable. Cette route est mutante mais non souveraine et non canonique.

Request

```
{
  "source": "session",
  "content_ref": "session:current",
  "target": "TRAINING_MEMORY",
  "reason": "useful_learning_candidate"
}
```

Response minimale attendue

```
{
  "id": "mem_candidate_001",
  "status": "CANDIDATE",
  "validation_required": "HUMAN_OR_PROCESS",
  "can_promote": false
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_create_memory_candidate.py
- tests/test_memory_candidate_no_auto_promote.py

Boundaries et interdits

- Créer directement PROMOTED_TO_CANON
- Écrire Graphiti
- Écrire canonical memory

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.24 POST /api/workcenter/memory/promotion/request

Version : V2

Page cockpit : Memory Governance

Ticket DEV : [V2] [MEMORY] [API] Demande promotion mémoire

Mode : VALIDATION_REQUEST

Objectif

Créer une demande de promotion vers training, operational, domain, proof ou canon. La route crée une validation request, pas une promotion finale.

Request

```
{
  "memory_candidate_id": "mem_candidate_001",
  "target": "CANONICAL_MEMORY",
  "reason": "validated doctrine candidate"
}
```

Response minimale attendue

```
{
  "validation_request_id": "val_001",
  "status": "PENDING",
  "required_actor": "human",
  "promotion_executed": false
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_memory_promotion_request.py

Boundaries et interdits

- Exécuter promotion immédiate
- Promouvoir canon sans validation humaine

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.25 GET /api/workcenter/validation/human

Version : V2

Page cockpit : Human Validation

Ticket DEV : [V2] [VALIDATION] [API] Queue validation humaine

Mode : READONLY

Objectif

Retourner les validations humaines en attente, approuvées, rejetées, expirées ou bloquées.

Request

```
No request body.
```

Response minimale attendue

```
{
  "items": [
    {
      "id": "val_001",
      "type": "MEMORY_CANON_PROMOTION",
      "status": "PENDING",
      "required_actor": "human",
      "risk_level": "HIGH",
      "evidence_refs": []
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_human_validation.py

Boundaries et interdits

- Approbation anonyme
- Auto approve

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.26 GET /api/workcenter/validation/human/{validation_id}

Version : V2

Page cockpit : Human Validation

Ticket DEV : [V2] [VALIDATION] [API] Détail validation humaine

Mode : READONLY

Objectif

Retourner le dossier complet d une validation avec sujet, source, preuves, commentaires et historique.

Request

```
No request body.
```

Response minimale attendue

```
{
  "id": "val_001",
  "subject_id": "mem_candidate_001",
  "status": "PENDING",
  "evidence_refs": [],
  "comments": [],
  "history": []
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_validation_detail.py

Boundaries et interdits

- Masquer le sujet
- Masquer les preuves manquantes

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.27 POST /api/workcenter/validation/request

Version : V2

Page cockpit : Human Validation

Ticket DEV : [V2] [VALIDATION] [API] Créer validation request

Mode : VALIDATION_REQUEST

Objectif

Créer une demande de validation structurée depuis une page cockpit. Cette route ne valide rien elle-même.

Request

```
{
  "type": "FREEZE_PUBLICATION",
  "subject_id": "freeze_candidate_001",
  "reason": "ready_for_review"
}
```

Response minimale attendue

```
{
  "id": "val_002",
  "status": "PENDING",
  "approval_executed": false
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_create_validation_request.py

Boundaries et interdits

- Créer APPROVED directement
- Omettre required_actor

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.28 POST /api/workcenter/validation/human/{validation_id}/approve

Version : V2

Page cockpit : Human Validation

Ticket DEV : [V2] [VALIDATION] [API] Approve validation

Mode : HUMAN_APPROVAL_RECORD

Objectif

Enregistrer l approbation humaine avec commentaire obligatoire pour les actions critiques. La route ne doit exécuter l action cible que si le workflow séparé est prévu.

Request

```
{
  "actor": "etienne_or_authorized_user",
  "comment": "reviewed evidence and approved",
  "evidence_checked": true
}
```

Response minimale attendue

```
{
  "id": "val_001",
  "status": "APPROVED",
  "action_executed": false,
  "next_step": "CALL_DOMAIN_WORKFLOW_OR_PROMOTION_EXECUTOR"
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_validation_approve_requires_comment.py

Boundaries et interdits

- Approbation sans commentaire pour HIGH risk
- Exécuter action sensible silencieusement

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.29 POST /api/workcenter/validation/human/{validation_id}/reject

Version : V2

Page cockpit : Human Validation

Ticket DEV : [V2] [VALIDATION] [API] Reject validation

Mode : HUMAN_REJECTION_RECORD

Objectif

Enregistrer le rejet humain avec justification obligatoire.

Request

```
{
  "actor": "authorized_user",
  "comment": "insufficient evidence"
}
```

Response minimale attendue

```
{
  "id": "val_001",
  "status": "REJECTED",
  "action_executed": false
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_validation_reject.py

Boundaries et interdits

- Rejet sans commentaire
- Suppression du dossier de validation

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.30 POST /api/workcenter/validation/human/{validation_id}/changes-requested

Version : V2

Page cockpit : Human Validation

Ticket DEV : [V2] [VALIDATION] [API] Changes requested

Mode : HUMAN_REVIEW_RECORD

Objectif

Demander une correction ou un complément de preuve sans approuver ni rejeter définitivement.

Request

```
{
  "actor": "authorized_user",
  "comment": "add proof report and freeze manifest"
}
```

Response minimale attendue

```
{
  "id": "val_001",
  "status": "CHANGES_REQUESTED",
  "action_executed": false
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_validation_changes_requested.py

Boundaries et interdits

- Transformer changes_requested en approved

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.31 GET /api/workcenter/validation/process

Version : V2

Page cockpit : Process Validation

Ticket DEV : [V2] [PROCESS] [API] Queue process validation

Mode : READONLY

Objectif

Retourner les validations par process, test, runtime, freeze, code ou document.

Request

```
No request body.
```

Response minimale attendue

```
{
  "items": [
    {
      "id": "proc_001",
      "target": "lean_build",
      "status": "CONFIRMED_BY_TEST",
      "evidence": "lake build completed successfully"
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_process_validation.py

Boundaries et interdits

- Surclasser un statut
- Masquer ENV_BLOCKED

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.32 GET /api/workcenter/validation/process/{process_id}

Version : V2

Page cockpit : Process Validation

Ticket DEV : [V2] [PROCESS] [API] Détail process validation

Mode : READONLY

Objectif

Retourner le détail d une validation process: commande, sortie, fichier preuve, timestamp, statut et limites.

Request

```
No request body.
```

Response minimale attendue

```
{
  "id": "proc_001",
  "command": "lake build",
  "status": "CONFIRMED_BY_TEST",
  "output_ref": "proofs/lean/build_output.txt",
  "limitations": []
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_process_validation_detail.py

Boundaries et interdits

- Présenter un résultat documenté comme exécuté

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.33 GET /api/workcenter/session/current

Version : V2

Page cockpit : Session Ledger

Ticket DEV : [V2] [SESSION] [API] Session actuelle

Mode : READONLY

Objectif

Retourner la session cockpit courante, ses événements, mémoires candidates, claims à valider, fichiers touchés et risques.

Request

```
No request body.
```

Response minimale attendue

```
{
  "session_id": "current",
  "status": "OPEN",
  "memory_candidates": [],
  "claims_to_validate": [],
  "actions_decided": [],
  "risks_and_blockers": []
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_current_session.py

Boundaries et interdits

- Écrire une mémoire durable
- Clôturer automatiquement

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.34 GET /api/workcenter/sessions

Version : V2

Page cockpit : Session Ledger

Ticket DEV : [V2] [SESSION] [API] Liste sessions

Mode : READONLY

Objectif

Lister les sessions connues avec statut, date, domaine, preuves, freezes et validations.

Request

```
Query parameters: status optional, domain_id optional, limit optional.
```

Response minimale attendue

```
{
  "items": [
    {
      "session_id": "sess_001",
      "status": "SESSION_SCRATCH",
      "domain_id": "bank",
      "created_at": "2026-06-17T12:00:00Z"
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_sessions.py

Boundaries et interdits

- Omettre status
- Mélanger sessions legacy et actives sans truth_status

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.35 GET /api/workcenter/sessions/{session_id}

Version : V2

Page cockpit : Session Ledger

Ticket DEV : [V2] [SESSION] [API] Détail session

Mode : READONLY

Objectif

Retourner la session complète avec timeline, events, décisions, preuves, memory candidates et outputs de clôture.

Request

```
No request body.
```

Response minimale attendue

```
{
  "session_id": "sess_001",
  "timeline": [],
  "decisions": [],
  "proof_refs": [],
  "outputs": {}
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_session_detail.py

Boundaries et interdits

- Masquer les erreurs de session
- Masquer les fichiers touchés

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.36 POST /api/workcenter/session/close/dry-run

Version : V2

Page cockpit : Session Ledger

Ticket DEV : [V2] [SESSION] [API] Clôture dry-run

Mode : DRY_RUN

Objectif

Préparer les outputs de clôture sans écrire durablement ni promouvoir.

Request

```
{
  "session_id": "sess_001",
  "include_memory_candidates": true,
  "include_next_commands": true
}
```

Response minimale attendue

```
{
  "SESSION_RECAP.md": "prepared",
  "MEMORY_CANDIDATES.json": "prepared",
  "CLAIMS_TO_VALIDATE.md": "prepared",
  "ACTIONS_DECIDED.md": "prepared",
  "TESTS_RUN.md": "prepared",
  "FILES_TOUCHED.md": "prepared",
  "RISKS_AND_BLOCKERS.md": "prepared",
  "NEXT_COMMANDS.md": "prepared",
  "dry_run": true
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_session_close_dry_run.py

Boundaries et interdits

- Promouvoir memory candidates
- Créer freeze final automatique

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.37 GET /api/workcenter/proofs

Version : V3

Page cockpit : Proof & Replay Lab

Ticket DEV : [V3] [PROOF] [API] Proofs summary

Mode : READONLY

Objectif

Retourner la synthèse des preuves: Lean, 57 théorèmes, TLA/TLC, verify_all, verify_decision, Merkle, RFC3161, adversarial tests, proofkits, freezes, latest decisions.

Request

```
No request body.
```

Response minimale attendue

```
{
  "summary": {
    "lean_theorem_count": 57,
    "lean_status": "CONFIRMED_BY_TEST",
    "tla_status": "PRESENT_NOT_RUN",
    "rfc3161_status": "PRESENT_NOT_VERIFIED_IN_CURRENT_AUDIT"
  },
  "items": []
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_proofs.py

Boundaries et interdits

- Afficher PRESENT_NOT_RUN comme PROVEN_BY_TEST
- Masquer warnings

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.38 GET /api/workcenter/proofs/{proof_id}

Version : V3

Page cockpit : Proof & Replay Lab

Ticket DEV : [V3] [PROOF] [API] Détail proof

Mode : READONLY

Objectif

Retourner la fiche d une preuve, sa source, son statut, sa dernière exécution, ses limites et ses artefacts.

Request

```
No request body.
```

Response minimale attendue

```
{
  "proof_id": "lean_basic",
  "type": "Lean",
  "status": "CONFIRMED_BY_TEST",
  "last_run": "2026-06-17",
  "artifacts": [],
  "limitations": []
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_proof_detail.py

Boundaries et interdits

- Omettre limitation
- Inventer une exécution

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.39 POST /api/workcenter/proofs/run/prepare

Version : V3

Page cockpit : Proof & Replay Lab

Ticket DEV : [V3] [PROOF] [API] Préparer proof run

Mode : PREPARE_COMMAND

Objectif

Préparer une commande de proof run en mode copy-only ou validation-required. Cette route ne doit pas exécuter la preuve sans workflow prévu.

Request

```
{
  "proof_type": "lean",
  "command": "lake build",
  "mode": "COPY_ONLY"
}
```

Response minimale attendue

```
{
  "prepared": true,
  "command": "cd proofs/lean && lake build",
  "execution_started": false,
  "validation_required": false
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_prepare_proof_run.py

Boundaries et interdits

- Exécuter une commande destructive
- Déclarer PASS avant run

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.40 GET /api/workcenter/replay/decisions

Version : V3

Page cockpit : Proof & Replay Lab

Ticket DEV : [V3] [REPLAY] [API] Replays disponibles

Mode : READONLY

Objectif

Lister les décisions rejouables avec domaine, gate, severity, reason_code, artefacts et statut de vérité.

Request

```
No request body.
```

Response minimale attendue

```
{
  "items": [
    {
      "decision_id": "dec_001",
      "domain": "bank",
      "gate": "ALLOW",
      "severity": "S1",
      "truth_status": "CONFIRMED_BY_FREEZE",
      "replay_available": true
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_replay_decisions.py

Boundaries et interdits

- Recalculer la décision
- Créer un replay sans source

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.41 GET /api/workcenter/replay/decisions/{decision_id}

Version : V3

Page cockpit : Proof & Replay Lab

Ticket DEV : [V3] [REPLAY] [API] Détail replay

Mode : READONLY

Objectif

Retourner le replay détaillé d une décision: payload, votes, aggregate, Kernel response, evidence_refs, attestation_ref, logs et limites.

Request

```
No request body.
```

Response minimale attendue

```
{
  "decision_id": "dec_001",
  "steps": [
    "payload",
    "agent_votes",
    "domain_aggregate",
    "kernel_decision",
    "proof_refs"
  ],
  "mutates_state": false
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_replay_detail.py

Boundaries et interdits

- Modifier l état en rejouant
- Masquer une étape manquante

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.42 GET /api/workcenter/freezes

Version : V3

Page cockpit : Freeze Factory

Ticket DEV : [V3] [FREEZE] [API] Freezes

Mode : READONLY

Objectif

Lister les freezes avec type, statut, manifest, hash, smoke, ledger entry et limitations.

Request

```
No request body.
```

Response minimale attendue

```
{
  "items": [
    {
      "freeze_id": "freeze_001",
      "type": "WORKCENTER_FREEZE",
      "status": "FROZEN",
      "manifest": "MANIFEST_SHA256.json",
      "truth_status": "CONFIRMED_BY_FREEZE"
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_freezes.py

Boundaries et interdits

- Afficher un freeze sans manifest comme complet
- Inclure secrets

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.43 GET /api/workcenter/freezes/{freeze_id}

Version : V3

Page cockpit : Freeze Factory

Ticket DEV : [V3] [FREEZE] [API] Détail freeze

Mode : READONLY

Objectif

Retourner un freeze complet: README, manifest, git_status, test_output, runtime_probe, openapi_snapshot, hash_report, registry_snapshot.

Request

```
No request body.
```

Response minimale attendue

```
{
  "freeze_id": "freeze_001",
  "files": [
    "README_FREEZE.md",
    "MANIFEST_SHA256.json",
    "git_status.txt",
    "test_output.txt",
    "runtime_probe.txt",
    "openapi_snapshot.json",
    "hash_report.json",
    "registry_snapshot.json"
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_freeze_detail.py

Boundaries et interdits

- Masquer git dirty
- Masquer test fail

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.44 POST /api/workcenter/freezes/candidate

Version : V3

Page cockpit : Freeze Factory

Ticket DEV : [V3] [FREEZE] [API] Préparer freeze candidate

Mode : FREEZE_CANDIDATE_REQUEST

Objectif

Créer un freeze candidate à valider, sans exécuter le freeze final.

Request

```
{
  "scope": "WORKCENTER_FREEZE",
  "include": [
    "registries",
    "routes",
    "tests"
  ],
  "reason": "checkpoint before implementation"
}
```

Response minimale attendue

```
{
  "freeze_candidate_id": "freeze_candidate_001",
  "status": "VALIDATION_REQUIRED",
  "freeze_executed": false
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_prepare_freeze_candidate.py

Boundaries et interdits

- Freeze automatique sans validation
- Freeze sans manifest
- Freeze avec secrets

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.45 GET /api/workcenter/freezes/{freeze_id}/manifest

Version : V3

Page cockpit : Freeze Factory

Ticket DEV : [V3] [FREEZE] [API] Lire manifest

Mode : READONLY

Objectif

Retourner le manifest SHA256 d un freeze sous forme lisible par l UI.

Request

```
No request body.
```

Response minimale attendue

```
{
  "freeze_id": "freeze_001",
  "manifest_status": "AVAILABLE",
  "items": [
    {
      "path": "README_FREEZE.md",
      "sha256": "..."
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_freeze_manifest.py

Boundaries et interdits

- Modifier manifest
- Recalculer en silence

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.46 GET /api/workcenter/freezes/compare

Version : V3

Page cockpit : Freeze Factory

Ticket DEV : [V3] [FREEZE] [API] Comparer freezes

Mode : READONLY

Objectif

Comparer deux freezes par fichiers, hash, routes, registries, tests et statuts.

Request

```
Query parameters: left_freeze_id, right_freeze_id.
```

Response minimale attendue

```
{
  "left": "freeze_001",
  "right": "freeze_002",
  "changed_files": [],
  "added_files": [],
  "removed_files": [],
  "status": "COMPARISON_READY"
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_compare_freezes.py

Boundaries et interdits

- Écrire dans les freezes
- Masquer fichiers supprimés

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.47 GET /api/workcenter/brody/training/modules

Version : V4

Page cockpit : Brody Education

Ticket DEV : [V4] [BRODY] [API] Modules training

Mode : READONLY

Objectif

Lister les modules T1 à T20 de Brody Education avec objectif, fichiers à lire, ressources, exercice, quiz, commande test, critère succès, statut, validation et actions interdites.

Request

```
No request body.
```

Response minimale attendue

```
{
  "items": [
    {
      "id": "T1",
      "title": "Comprendre Obsidia / X108",
      "mode": "READONLY_LEARNING",
      "forbidden_actions": [
        "ACT",
        "commit",
        "kernel_mutation"
      ]
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_brody_training_modules.py

Boundaries et interdits

- Module sans forbidden_actions
- Brody write

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.48 GET /api/workcenter/brody/training/modules/{module_id}

Version : V4

Page cockpit : Brody Education

Ticket DEV : [V4] [BRODY] [API] Détail module

Mode : READONLY

Objectif

Retourner le contenu structuré d un module Brody.

Request

```
No request body.
```

Response minimale attendue

```
{
  "id": "T1",
  "objective": "Comprendre Obsidia / X108",
  "files_to_read": [],
  "exercise": {},
  "quiz": {},
  "success_criteria": []
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_brody_training_module_detail.py

Boundaries et interdits

- Inclure secrets
- Inclure action souveraine

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.49 POST /api/workcenter/brody/training/quiz/dry-run

Version : V4

Page cockpit : Brody Education

Ticket DEV : [V4] [BRODY] [API] Quiz dry-run

Mode : DRY_RUN

Objectif

Corriger un quiz Brody en dry-run et produire un résultat pédagogique sans écrire de mémoire canonique.

Request

```
{
  "module_id": "T1",
  "answers": {
    "q1": "KX108_ONLY"
  }
}
```

Response minimale attendue

```
{
  "score": 1,
  "passed": true,
  "memory_candidate_created": false,
  "can_decide": false
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_brody_quiz_dry_run.py

Boundaries et interdits

- Créer mémoire durable
- Attribuer autorité à Brody

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.50 GET /api/workcenter/resources

Version : V4

Page cockpit : Resource Library

Ticket DEV : [V4] [RESOURCE] [API] Ressources

Mode : READONLY

Objectif

Lister les ressources: README, glossaire, recheck reports, audits, freezes, prompts, pitchs, schemas, payloads, scripts, proof outputs, training modules, operator guides, error catalog, diagrams, multimodal extracts.

Request

```
No request body.
```

Response minimale attendue

```
{
  "items": [
    {
      "id": "readme_v2",
      "title": "README OBSIDIA X108",
      "type": "README",
      "truth_level": "ACTIVE_REFERENCE",
      "legacy_or_active": "active"
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_resources.py

Boundaries et interdits

- Source sans statut
- Confondre legacy et active

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.51 GET /api/workcenter/resources/{resource_id}

Version : V4

Page cockpit : Resource Library

Ticket DEV : [V4] [RESOURCE] [API] Détail ressource

Mode : READONLY

Objectif

Retourner les métadonnées d une ressource: chemin, scope, status, last_verified, used_by, linked_modules, risk, legacy_or_active.

Request

```
No request body.
```

Response minimale attendue

```
{
  "id": "readme_v2",
  "path": "README_OBSIDIA_X108_GITHUB_V2.md",
  "status": "ACTIVE",
  "used_by": [
    "Dashboard",
    "Resource Library"
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_resource_detail.py

Boundaries et interdits

- Lire contenu secret
- Masquer obsolescence

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.52 GET /api/workcenter/terrain/domains

Version : V4

Page cockpit : Terrain Branching

Ticket DEV : [V4] [TERRAIN] [API] Domaines terrain

Mode : READONLY

Objectif

Retourner les domaines terrain avec status, payload_schema, example_payloads, connector, adapter, routes, risk_metrics, last_runtime, last_freeze, last_replay, last_decision, allowed_modes, forbidden_modes.

Request

```
No request body.
```

Response minimale attendue

```
{
  "items": [
    {
      "domain_id": "bank",
      "status": "CONFIRMED_BY_FREEZE",
      "allowed_modes": [
        "SIMULATED",
        "RECORDED",
        "REPLAY",
        "READONLY_LIVE"
      ],
      "forbidden_modes": [
        "DIRECT_ACT",
        "DOMAIN_DECIDES_ALONE"
      ]
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_terrain_domains.py

Boundaries et interdits

- REAL_ACTION_WITHOUT_X108
- UNVERIFIED_LIVE

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.53 GET /api/workcenter/terrain/branches

Version : V4

Page cockpit : Terrain Branching

Ticket DEV : [V4] [TERRAIN] [API] Branches

Mode : READONLY

Objectif

Lister les branches terrain candidates, actives, merged, rejected ou blocked.

Request

```
No request body.
```

Response minimale attendue

```
{
  "items": [
    {
      "branch_id": "terrain_bank_v1",
      "parent_branch_id": "main",
      "status": "CANDIDATE",
      "domain_id": "bank",
      "freeze_required": true
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_terrain_branches.py

Boundaries et interdits

- Fusion automatique sans validation
- Action réelle directe

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.54 POST /api/workcenter/terrain/branches/candidate

Version : V4

Page cockpit : Terrain Branching

Ticket DEV : [V4] [TERRAIN] [API] Créer branche candidate

Mode : VALIDATION_REQUEST

Objectif

Préparer une branche terrain candidate à valider.

Request

```
{
  "domain_id": "bank",
  "parent_branch_id": "main",
  "purpose": "test new threshold",
  "risk_metrics": [
    "false_positive_rate",
    "latency_ms"
  ]
}
```

Response minimale attendue

```
{
  "branch_id": "terrain_candidate_001",
  "status": "VALIDATION_REQUIRED",
  "activated": false
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_create_terrain_branch_candidate.py

Boundaries et interdits

- Activation live sans validation
- Domaine qui décide seul

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.55 GET /api/workcenter/agents

Version : V4

Page cockpit : Agent Center

Ticket DEV : [V4] [AGENT] [API] Agents

Mode : READONLY

Objectif

Lister les agents domaine, meta-agents, agents Brody et agents non souverains avec rôle et boundaries.

Request

```
No request body.
```

Response minimale attendue

```
{
  "items": [
    {
      "agent_id": "bank_risk_agent",
      "domain": "bank",
      "role": "PERIPHERAL_ADVISORY",
      "can_decide": false
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_agents.py

Boundaries et interdits

- Agent can_decide true hors Kernel
- Agent emit_act true

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.56 GET /api/workcenter/knowledge/candidates

Version : V4

Page cockpit : Knowledge Center

Ticket DEV : [V4] [KNOWLEDGE] [API] Knowledge candidates

Mode : READONLY

Objectif

Lister les connaissances candidates issues de sessions, audits, multimodal, Brody ou ressources.

Request

```
No request body.
```

Response minimale attendue

```
{
  "items": [
    {
      "id": "know_candidate_001",
      "source": "audit",
      "status": "NEEDS_HUMAN_REVIEW",
      "target": "TRAINING_MEMORY"
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_knowledge_candidates.py

Boundaries et interdits

- Canonisation automatique
- Diffusion aux agents opérationnels sans validation

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.57 GET /api/workcenter/sandbox/policy

Version : V5

Page cockpit : Coding Sandbox

Ticket DEV : [V5] [SANDBOX] [API] Policy sandbox

Mode : READONLY

Objectif

Retourner la policy du sandbox: niveaux, zones, actions autorisées, actions interdites, protocole Kernel.

Request

```
No request body.
```

Response minimale attendue

```
{
  "levels": [
    "READ_ONLY_LEARNING",
    "TEST_ONLY",
    "PATCH_PROPOSAL",
    "LOCAL_SANDBOX_WRITE",
    "HUMAN_APPROVED_WRITE",
    "KERNEL_TOUCH_PROTOCOL_REQUIRED"
  ],
  "forbidden": [
    "edit_kernel_without_protocol",
    "edit_env",
    "delete_freeze",
    "commit_directly",
    "merge_directly",
    "write_graphiti",
    "emit_act"
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_sandbox_policy.py

Boundaries et interdits

- Autoriser kernel edit sans protocol
- Autoriser env edit

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.58 GET /api/workcenter/sandbox/files/tree

Version : V5

Page cockpit : Coding Sandbox

Ticket DEV : [V5] [SANDBOX] [API] Arborescence readonly

Mode : READONLY

Objectif

Retourner une arborescence de fichiers consultable avec classification safe_to_read, safe_to_test, safe_to_patch, requires_review, kernel_protected, secrets_forbidden, legacy_unknown.

Request

```
No request body.
```

Response minimale attendue

```
{
  "items": [
    {
      "path": "apps/obsidia_api/main.py",
      "zone": "safe_to_read"
    },
    {
      "path": "server.kernel.sealed.cjs",
      "zone": "kernel_protected"
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_sandbox_files_tree.py

Boundaries et interdits

- Afficher secrets
- Permettre édition kernel_protected

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.59 GET /api/workcenter/sandbox/files/read

Version : V5

Page cockpit : Coding Sandbox

Ticket DEV : [V5] [SANDBOX] [API] Lire fichier readonly

Mode : READONLY

Objectif

Lire un fichier classé safe_to_read ou safe_to_test.

Request

```
Query parameters: path.
```

Response minimale attendue

```
{
  "path": "apps/obsidia_api/main.py",
  "zone": "safe_to_read",
  "content": "...",
  "editable": false
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_sandbox_file_read.py

Boundaries et interdits

- Lire .env
- Lire secrets
- Lire fichier kernel protégé sans protocole

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.60 POST /api/workcenter/sandbox/patch-proposals

Version : V5

Page cockpit : Coding Sandbox

Ticket DEV : [V5] [SANDBOX] [API] Créer patch proposal

Mode : PATCH_PROPOSAL

Objectif

Créer une proposition de patch, sans application automatique.

Request

```
{
  "target_path": "apps/obsidia_api/routes/workcenter_status.py",
  "diff": "--- a/...",
  "reason": "add boundary field"
}
```

Response minimale attendue

```
{
  "patch_id": "patch_001",
  "status": "NEEDS_HUMAN_REVIEW",
  "applied": false
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_sandbox_patch_proposal.py

Boundaries et interdits

- Appliquer patch automatiquement
- Commit direct
- Modifier Kernel sans protocol

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.61 POST /api/workcenter/sandbox/test-proposals

Version : V5

Page cockpit : Coding Sandbox

Ticket DEV : [V5] [SANDBOX] [API] Test proposal

Mode : TEST_PROPOSAL

Objectif

Créer une proposition de test ou commande test à valider ou copier.

Request

```
{
  "test_file": "tests/test_workcenter_status.py",
  "command": "pytest tests/test_workcenter_status.py -v"
}
```

Response minimale attendue

```
{
  "test_proposal_id": "test_prop_001",
  "status": "PROPOSED",
  "executed": false
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_sandbox_test_proposal.py

Boundaries et interdits

- Exécution destructive
- Exécution réseau non autorisée

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.62 GET /api/workcenter/multimodal/assets

Version : V5

Page cockpit : Multimodal Workbench

Ticket DEV : [V5] [MULTIMODAL] [API] Assets multimodaux

Mode : READONLY

Objectif

Lister les assets multimodaux: image, audio, vidéo, PDF, captures écran, logs, schémas, slides, terminal.

Request

```
No request body.
```

Response minimale attendue

```
{
  "items": [
    {
      "asset_id": "asset_001",
      "type": "PDF",
      "status": "RAW_CAPTURE",
      "linked_resources": []
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_multimodal_assets.py

Boundaries et interdits

- Décider depuis un asset
- Écrire mémoire durable directement

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.63 POST /api/workcenter/multimodal/extractions/candidate

Version : V5

Page cockpit : Multimodal Workbench

Ticket DEV : [V5] [MULTIMODAL] [API] Créer extraction candidate

Mode : CANDIDATE_CREATE

Objectif

Créer une extraction candidate depuis un asset, utilisable pour validation mémoire ou ressource.

Request

```
{
  "asset_id": "asset_001",
  "extraction_type": "TEXT_SUMMARY",
  "target": "MEMORY_CANDIDATE"
}
```

Response minimale attendue

```
{
  "extraction_id": "extract_001",
  "status": "CANDIDATE",
  "memory_write": false
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_multimodal_extraction_candidate.py

Boundaries et interdits

- Promotion durable automatique
- Déclencher action

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.64 GET /api/workcenter/governance/policies

Version : V5

Page cockpit : Governance Policies

Ticket DEV : [V5] [POLICY] [API] Policies

Mode : READONLY

Objectif

Lister les policies: decision_authority, memory_promotion, freeze, proof_claim, brody_boundary, graphiti_readonly, kernel_change, sandbox, server_advisory, multimodal_ingestion.

Request

```
No request body.
```

Response minimale attendue

```
{
  "items": [
    {
      "policy_id": "decision_authority_policy",
      "status": "ACTIVE",
      "rule": "Only Kernel X108 can decide"
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_governance_policies.py

Boundaries et interdits

- Policy sans source
- Policy contredisant KX108_ONLY

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.65 GET /api/workcenter/governance/policies/{policy_id}

Version : V5

Page cockpit : Governance Policies

Ticket DEV : [V5] [POLICY] [API] Détail policy

Mode : READONLY

Objectif

Retourner règle, raison, composants concernés, actions autorisées, actions interdites, version, statut, source, dernier audit.

Request

```
No request body.
```

Response minimale attendue

```
{
  "policy_id": "kernel_change_policy",
  "allowed_actions": [
    "request_change"
  ],
  "forbidden_actions": [
    "direct_kernel_mutation_from_cockpit"
  ],
  "status": "ACTIVE"
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_governance_policy_detail.py

Boundaries et interdits

- Autoriser direct_kernel_mutation_from_cockpit

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.66 POST /api/workcenter/kernel/change-requests

Version : V5

Page cockpit : Kernel Change Protocol

Ticket DEV : [V5] [KERNEL_CHANGE] [API] Créer request changement Kernel

Mode : KERNEL_CHANGE_REQUEST_ONLY

Objectif

Créer une demande de changement Kernel. Cette route ne modifie pas le Kernel. Elle ouvre un dossier audité soumis au Kernel Change Protocol.

Request

```
{
  "scope": "kernel_contract",
  "reason": "documented evolution request",
  "files": [
    "server.kernel.sealed.cjs"
  ],
  "risk_level": "CRITICAL"
}
```

Response minimale attendue

```
{
  "change_request_id": "kcr_001",
  "status": "PENDING",
  "kernel_modified": false,
  "required_protocol": "KERNEL_CHANGE_PROTOCOL"
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_kernel_change_request_only.py

Boundaries et interdits

- Modifier le Kernel
- Créer patch appliqué
- Bypass proof/freeze

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.67 GET /api/workcenter/audit/events

Version : V5

Page cockpit : Audit Center

Ticket DEV : [V5] [AUDIT] [API] Audit events

Mode : READONLY

Objectif

Lister les événements auditables du cockpit: lectures, demandes, validations, proposals, preuves, freezes et warnings.

Request

```
Query parameters: actor optional, event_type optional, from optional, to optional, limit optional.
```

Response minimale attendue

```
{
  "items": [
    {
      "event_id": "evt_001",
      "event_type": "validation.requested",
      "actor": "cockpit",
      "timestamp": "2026-06-17T12:00:00Z",
      "trace_id": "trace_001"
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_workcenter_audit_events.py

Boundaries et interdits

- Événement sans actor
- Événement sans timestamp

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

8.68 GET /api/workcenter/acceptance/criteria

Version : V5

Page cockpit : Acceptance Matrix

Ticket DEV : [V5] [ACCEPTANCE] [API] Acceptance criteria

Mode : READONLY

Objectif

Retourner tous les critères d acceptation du cockpit par page, version, test et statut.

Request

```
No request body.
```

Response minimale attendue

```
{
  "items": [
    {
      "id": "acc_dashboard_001",
      "page": "Dashboard global",
      "status": "PENDING",
      "test": "tests/frontend/dashboard_global.spec.ts"
    }
  ]
}
```

Erreurs possibles

- 400 VALIDATION_ERROR
- 403 FORBIDDEN_BY_BOUNDARY
- 404 NOT_FOUND
- 409 STATUS_CONFLICT
- 500 INTERNAL_ERROR

Tests obligatoires

- tests/test_acceptance_criteria.py

Boundaries et interdits

- Marquer PASS sans test ou validation

Critere de reussite

La route est acceptee si elle retourne le schema attendu, conserve le statut de verite exact, expose les boundaries, journalise la trace, ne declenche aucune action interdite et possede un test automatise PASS.

9. Matrice de tests par famille

Foundation / Registries

- tests/test_workcenter_boundaries.py
- tests/test_workcenter_component_detail.py
- tests/test_workcenter_components.py
- tests/test_workcenter_metrics.py
- tests/test_workcenter_no_decision_flags.py
- tests/test_workcenter_registries_index.py
- tests/test_workcenter_route_detail.py
- tests/test_workcenter_routes_catalog.py
- tests/test_workcenter_status.py

Dashboard global

- tests/test_workcenter_dashboard.py
- tests/test_workcenter_open_risks.py
- tests/test_workcenter_recent_decisions.py

Authority Graph

- tests/test_workcenter_authority.py
- tests/test_workcenter_forbidden_actions.py

Server Advisory

- tests/test_workcenter_ports.py
- tests/test_workcenter_servers.py
- tests/test_workcenter_suggested_commands.py

Domain Center

- tests/test_workcenter_domain_detail.py
- tests/test_workcenter_domains.py

Routes Registry

- tests/test_workcenter_page_routes.py

Memory Governance

- tests/test_memory_candidate_no_auto_promote.py
- tests/test_memory_promotion_request.py
- tests/test_workcenter_create_memory_candidate.py
- tests/test_workcenter_memory_candidates.py
- tests/test_workcenter_memory_detail.py
- tests/test_workcenter_promoted_memory.py

Human Validation

- tests/test_create_validation_request.py
- tests/test_validation_approve_requires_comment.py
- tests/test_validation_changes_requested.py
- tests/test_validation_reject.py
- tests/test_workcenter_human_validation.py
- tests/test_workcenter_validation_detail.py

Process Validation

- tests/test_workcenter_process_validation.py
- tests/test_workcenter_process_validation_detail.py

Session Ledger

- tests/test_session_close_dry_run.py
- tests/test_workcenter_current_session.py
- tests/test_workcenter_session_detail.py
- tests/test_workcenter_sessions.py

Proof & Replay Lab

- tests/test_prepare_proof_run.py
- tests/test_workcenter_proof_detail.py
- tests/test_workcenter_proofs.py
- tests/test_workcenter_replay_decisions.py
- tests/test_workcenter_replay_detail.py

Freeze Factory

- tests/test_compare_freezes.py
- tests/test_prepare_freeze_candidate.py
- tests/test_workcenter_freeze_detail.py
- tests/test_workcenter_freeze_manifest.py
- tests/test_workcenter_freezes.py

Brody Education

- tests/test_brody_quiz_dry_run.py
- tests/test_brody_training_module_detail.py
- tests/test_brody_training_modules.py

Resource Library

- tests/test_workcenter_resource_detail.py
- tests/test_workcenter_resources.py

Terrain Branching

- tests/test_create_terrain_branch_candidate.py
- tests/test_terrain_branches.py
- tests/test_terrain_domains.py

Agent Center

- tests/test_workcenter_agents.py

Knowledge Center

- tests/test_workcenter_knowledge_candidates.py

Coding Sandbox

- tests/test_sandbox_file_read.py
- tests/test_sandbox_files_tree.py
- tests/test_sandbox_patch_proposal.py
- tests/test_sandbox_policy.py
- tests/test_sandbox_test_proposal.py

Multimodal Workbench

- tests/test_multimodal_assets.py
- tests/test_multimodal_extraction_candidate.py

Governance Policies

- tests/test_governance_policies.py

- tests/test_governance_policy_detail.py

Kernel Change Protocol

- tests/test_kernel_change_request_only.py

Audit Center

- tests/test_workcenter_audit_events.py

Acceptance Matrix

- tests/test_acceptance_criteria.py

10. Tests de boundaries transverses

Ces tests ne sont pas lies a une seule route. Ils doivent traverser tout le WorkCenter API.

- `tests/test_workcenter_never_decides.py` verifie que toutes les routes WorkCenter conservent allowed_to_decide=false.
- `tests/test_workcenter_no_emit_act.py` verifie que toutes les routes conservent emits_act=false.
- `tests/test_workcenter_no_kernel_mutation.py` verifie qu aucune route ne modifie directement le Kernel.
- `tests/test_workcenter_no_graphiti_write.py` verifie que Graphiti reste readonly.
- `tests/test_workcenter_no_auto_canon.py` verifie qu aucune route ne promeut automatiquement une memoire canonique.
- `tests/test_workcenter_truth_status_integrity.py` verifie qu un statut freeze, code ou documented only n est pas surclasse.
- `tests/test_workcenter_no_secret_leak.py` verifie qu aucune route ne renvoie .env, token, key, secret, password ou contenu interdit.
- `tests/test_workcenter_trace_required.py` verifie que toutes les routes renvoient trace_id ou equivalent auditable.
- `tests/test_workcenter_post_routes_are_requests.py` verifie que les routes POST creent des demandes ou candidates et non des actes reels.

11. Regles de nommage fichiers backend

Les routes WorkCenter doivent etre separees par domaine fonctionnel pour eviter un fichier unique illisible.

```
apps/obsidia_api/routes/
  workcenter_status.py
  workcenter_metrics.py
  workcenter_registries.py
  workcenter_components.py
  workcenter_routes.py
  workcenter_authority.py
  workcenter_dashboard.py
  workcenter_servers.py
  workcenter_domains.py
  workcenter_memory.py
  workcenter_validation.py
  workcenter_process_validation.py
  workcenter_session.py
  workcenter_proofs.py
  workcenter_replay.py
  workcenter_freezes.py
  workcenter_brody.py
  workcenter_resources.py
  workcenter_terrain.py
  workcenter_agents.py
  workcenter_knowledge.py
```

```
  workcenter_sandbox.py
  workcenter_multimodal.py
  workcenter_governance.py
  workcenter_kernel_change.py
  workcenter_audit.py
  workcenter_acceptance.py
```

12. Regles de nommage schemas

```
apps/obsidia_api/schemas/workcenter/
  status.py
  metrics.py
  registry.py
  component.py
  route.py
  authority.py
  dashboard.py
  server.py
  domain.py
  memory.py
  validation.py
  process_validation.py
  session.py
  proof.py
  replay.py
  freeze.py
  brody.py
  resource.py
  terrain.py
  agent.py
  knowledge.py
  sandbox.py
  multimodal.py
  governance.py
  kernel_change.py
  audit.py
  acceptance.py
```

13. Regles de nommage tests

```
tests/workcenter/
  test_status.py
  test_metrics.py
  test_registries.py
  test_components.py
  test_routes.py
  test_authority.py
  test_dashboard.py
  test_servers.py
  test_domains.py
  test_memory.py
  test_validation.py
  test_process_validation.py
  test_session.py
  test_proofs.py
  test_replay.py
  test_freezes.py
```

```
test_brody.py
test_resources.py
test_terrain.py
test_agents.py
test_knowledge.py
test_sandbox.py
test_multimodal.py
test_governance.py
test_kernel_change.py
test_audit.py
test_acceptance.py
test_boundaries_global.py
```

14. Definition of Done du Contrat API

Le contrat API WorkCenter est considere pret pour build seulement si les points suivants sont coches.

- Toutes les routes V0 sont listees, schema inclus, tests inclus.
- Toutes les routes V1 sont reliees aux pages Dashboard, Authority, Server Advisory, Domain Center, Routes Registry.
- Toutes les routes V2 couvrent Memory Governance, Human Validation, Process Validation, Session Ledger.
- Toutes les routes V3 couvrent Proof & Replay Lab et Freeze Factory.
- Toutes les routes V4 couvrent Brody Education, Resource Library, Terrain Branching, Agent Center, Knowledge Center.
- Toutes les routes V5 couvrent Coding Sandbox, Multimodal Workbench, Governance Policies, Kernel Change Protocol, Audit Center, Acceptance Matrix.
- Chaque POST est classe comme request, candidate, dry-run, validation record, proposal ou prepare command.
- Aucune route ne declare une decision souveraine cockpit.
- Aucune route ne modifie directement le Kernel.
- Aucune route necrit Graphiti.
- Aucune route ne promeut la memoire canonique sans validation humaine.
- Les erreurs standard sont presentes.
- Les tests transverses boundaries sont definis.
- Les statuts de verite ne peuvent pas etre surclasses sans preuve.

15. Synthese pour developpeur

Le developpeur doit commencer par V0. Il ne doit pas commencer par une page UI avancee. La premiere tache est de creer les registries, les schemas et les routes readonly de base. Ensuite seulement, le Dashboard et l Authority Graph peuvent etre construits. Les routes V2 a V5 doivent etre ajoutees progressivement, en gardant la meme enveloppe de reponse et les memes tests de boundary.

```
Ordre recommande:
1. Schemas de base WorkCenter.
2. Routes V0 status, metrics, registries, components, routes, boundaries.
3. Tests V0.
4. Routes V1 dashboard, authority, servers, domains.
5. UI V1.
6. Routes V2 memory, validation, process, session.
7. UI V2.
8. Routes V3 proofs, replay, freezes.
9. UI V3.
10. Routes V4 Brody, resources, terrain, agents, knowledge.
11. UI V4.
12. Routes V5 sandbox, multimodal, governance, kernel change request, audit, acceptance.
13. UI V5.
14. Tests transverses boundaries.
```

15. Freeze WorkCenter.